Roll No. .....

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31N0405

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B. TECH. III Sem Main/Back Exam
Academic Session : 2024-25
Computer Science & Engineering
(3CS4-05) - Software Engineering
3AD4-05, 3AM4-05, 3AI4-05, 3CM4-05, 3CY4-05, 3CD4-05, 3CS4-05, 3IT4-05, 3IC4-05

Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours] **[Max. Marks : 70**

[Min. Passing Marks :

Instructions to Candidates :

Part-A : Short answer questions (up to 25 words) 10 * 2 marks = 20marks. All ten questions are compulsory.

Part-B : Analytical/Problem Solving questions 5 * 4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part-C : Descriptive/Analytical/Problem Solving questions 3 &* 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly.

Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination.

(Mentioned in form No. 205)

1_____ 2_____

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Part-A

10x2=20

Q.1 Define SDLC with its various phases. [2]

Q.2 Differentiate between verification and validation. [2]

Q.3 What do you mean by Metrics ? [2]

Q.4 Define SRS. [2]

Q.5 What are the characteristics of SRS ? [2]

Q.6 Define Software Process. [2]

Q.7 What is data dictionary ? [2]

Q.8 What is meant by Information Flow Continuity ? [2]

Q.9 What is Data Modeling ? [2]

Q.10 Define Finite State Machine (FSA) models. [2]

Part-B

5x4=20

Q.1 What are the characteristics of a good design ? Describe different types of coupling and cohesion. How design evaluation is performed ? [4]

Q.2 Discuss Object Oriented Analysis (OOA) and Modeling in details. [4]

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Q.3 What do you understand data dictionary ? When and how it is used ? [4]

Q.4 Write out the reasons for the failure of Waterfall Model. [4]

Q.5 What are Size Metrics ? How (FP) Function Point Metric advantages over LoC Metrics ? [4]

Q.6 Differentiate between a Software Product and a Software Process. [4]

Q.7 Discuss data and Control Flow Diagram in details. [4]

Section-C

3x10=30

Q.1 What is SDLC ? Explain Spiral Model of Software Development Life Cycle. [10]

Q.2 What is Modularity ? State its importance and explain Coupling and Cohesion. [10]

Q.3 What is the purpose of Data Flow Diagram ? What are the notations used for the same. Explain by constructing a context Flow Diagram Level - 0 DFD and Level - 1 DFD for a Library Management System. [10]

Q.4 Discuss about COCOMO II Model for Software Estimation. [10]

Q.5 What is SRS ? Write the characteristics of goods SRS document. [10]

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